Jason Duran Dutton

Master Resume

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Professional Profile

Technical practitioner whose work spans industrial electrical maintenance, process operations, steel fabrication, construction, automotive operations, and independent software and research projects. Experience includes fault isolation, equipment repair, crew coordination, technical documentation, and translating system behavior into practical diagnostic steps. Current public work includes Python modeling tools, synthetic measurement diagnostics, interactive visualizations, and research manuscripts. Currently enrolled in the BSIT program at American InterContinental University, with a long-term goal of doctoral study in computational physics.

Core Capabilities

Electrical and mechanical troubleshooting; industrial maintenance; motors, controls, conveyors and pumping systems; drawings and wiring schematics; MIG and arc welding; fabrication and field coordination; supervision and training; inspection documentation; Python; numerical modeling; Git/GitHub; LaTeX; technical writing and interactive visualization.

Industrial and Operational Experience

Underground Electrical Repairman and Outby Electrician

Paramont Contura / Alpha Metallurgical Resources | 2022-2025*

- Maintained and repaired electrical and mechanical infrastructure supporting underground operations, including beltlines, drives, controls and pumping equipment.
- Performed inspections and compliance documentation; diagnosed production-critical equipment failures and coordinated repair and recovery work.
- Worked with mining engineers on underground pump selection, installation and water-management requirements, including drainage, dust suppression and beltline protection.

Residential Construction

Self-employed | Began 2018; latest end date unconfirmed

- Undertook residential construction work as a self-employed practitioner. Individual projects, scope and completion records remain to be added to the master archive.

Shop and Field Fabricator Welder and Foreman

Dominion Steel Inc. | Fredericksburg, Virginia | 2016-2018

- Fabricated and welded structural steel to detail drawings; supervised and trained shop and field personnel.
- Coordinated materials transport, installation, crew allocation and production requirements with management, customers, suppliers and engineers.
- Worked on steel erection and structural retrofits in Northern Virginia, Southern Maryland and Washington, D.C.

Industrial and Operational Experience Continued

Pilot Plant and Process Operator

Minerals Refining Company | St. Charles, Virginia | 2015-2016

- Participated in construction and day-to-day operation of a pilot process for recovering ultrafine coal from preparation-plant wastewater using Virginia Tech-developed separation technology.
- Maintained and repaired mechanical and electrical process equipment; interpreted wiring diagrams and schematics and supported installation of process and electrical equipment.
- Kept electrical-equipment records and pre-shift/on-shift examination logs; supervised and trained personnel and supported mine-safety and workplace-safety requirements.

Later source documents describe this work as lead electrical integration and commissioning. The master record retains the earlier pilot/process title while the exact role and employer/site naming are reconciled.

Underground Electrician and Maintenance Foreman

Alpha Natural Resources / Dickenson Russell Coal Company | 2010-2015*

- Maintained, diagnosed and repaired underground and surface electrical, mechanical and diesel equipment at the Laurel Mountain and Cherokee mines.
- Troubleshot motors, controls, conveyors, pumps and electrical distribution systems; read prints and wiring diagrams and installed equipment.
- Supervised and trained personnel, performed MIG and arc welding, and maintained equipment records for federal and state mining requirements.

Automotive Experience

Sales Professional

Morgan McClure Chevrolet Inc. | Coeburn, Virginia | 2007-2010

- Managed customer relationships and vehicle sales transactions through completion, including financing coordination and finance/insurance-related steps in the sales process.
- Developed relationships with local lenders to broaden financing options, including work with customers facing financing challenges.

Inventory Control Manager

Deardoff Automotive Group / Continental Motorcars Inc. | Melbourne, Florida | 2002-2007

- Progressed from auto detailing and vehicle washing to inventory control and oversight of parts and detailing operations.
- Combined automotive work with college coursework; was offered advancement and testing opportunities through Ford Motor Company's Stars II system.

Selected Industrial Case Study

14CM15 continuous miner tram failure: the self-authored case study describes a controlled alternate-path test that separated drive failure from control and wiring hypotheses and isolated a failed left tram VFD. It illustrates variable isolation and diagnostic test design. The event date and independent repair records are not yet documented; no downtime savings are claimed.

Software and Computational Research

Independent Software and Research Projects

Public portfolio active in 2026 | github.com/JasonSEFIDEFI

Develops computational tools and documentation for geometric modeling, synthetic measurement diagnostics, numerical experimentation and technical explanation. Public repository dates establish the recorded portfolio activity; earlier start dates remain under documentation.

SEFI PY Engine

- Maintains Python source for geometric/evolution models, numerical workflows and visualization, with project documentation and test sources.
- Organizes computational implementations alongside explicit assumptions, reproducibility instructions and research limitations.

Microwave Calibration and Drift Diagnostic Workbench

- Implements a synthetic measurement-to-report workflow comparing independent fitting, conventional temporal tracking and a geometry-inspired tangent-tracking method.
- Includes CSV replay, separately calibrated alert thresholds, checks for ambiguous evidence and criteria for recovery acceptance.
- The archived validation record reports eleven passing tests. Its benchmark does not establish an advantage over conventional tracking; data are synthetic and no physical equipment was tested.

Vortex Research Console

- Provides a local interface for calculated classical field profiles, axisymmetric evolution, parameter exploration, saved frames and computational checkpoints.
- Documents finite-domain numerical limits, incomplete stability analysis and failed searches. Numerical candidates are not presented as derived observed particles or established physical theory.

SEFI QEC Stack

- Maintains modular Python software for stabilizer checks, syndrome processing, geometric representations and correction-workflow exploration.
- Contains test sources, Q# material and optional Azure Quantum integration code. Physical quantum-error suppression and executed hardware results are not established by the reviewed archive.

Interactive Double Slit and Waveform Visualizations

- Develops interactive interfaces for explaining interference and proposed geometric interpretations. The current educational explorer uses a standard reference probability model and distinguishes interpretive overlays from physical predictions.
- Earlier React/JavaScript prototypes preserve the development history; some are incomplete and are not represented as validated production applications.

Research Manuscripts and Technical Concepts

Author of research manuscripts and technical proposals exploring geometric field models, identity-space constructions, coupled dynamics, quantum-information concepts and acoustic visualization. Journal submission activity is in progress; acceptance and peer-reviewed publication have not been established in the reviewed evidence.

- Geometric Worldline Foundations of the Electron Field
- SEFI The Single Entity Field Interpretation and formal mathematical construction
- Dynamic Entity Field Integration A Coupled Lagrangian Construction Unifying SEFI Identity and GWFM Geometry
- Unified Geometric Field Theory from Worldline and Identity-Space Invariants
- Geometric Photonic Quantum Error Correction via SEFI GWFM and DEFI
- DNA as a Golden-Ratio Helical Worldline and displacement/warp geometry proposals
- Acoustic-Levitated Dynamic Voxel System for Volumetric Field Visualization: proposed control/illumination architecture; a built and validated hardware system is not established.

Education and Training

American InterContinental University (AIU)

Bachelor of Science in Information Technology (BSIT) | Currently enrolled; degree in progress

Long-term academic goal: a PhD in computational physics. Building computing foundations through the BSIT alongside independent Python research and further preparation in mathematics, numerical methods, and physics.

Mountain Empire Community College

Mining electrical and underground mining studies | 2009-2011

Brevard Community College

Computer Information Systems and Business Management coursework | 2002-2004

Earlier records associate this coursework with UCF. The precise institutional relationship remains unconfirmed; no college degree is claimed.

Castlewood High School

General high school diploma | 2002

Historical Qualifications

Previously reported Virginia mining qualifications: Underground General Miner; Qualified Gas Detector; Electrical Maintenance Foreman; Diesel Engine Repairman; Underground Electrical Repairman. Previously reported federal qualification: Underground and Surface Electrical Repairman. Renewal records and current validity have not been verified.

Previously reported American Heart Association training includes first aid, adult/child/infant CPR and AED. Welding and troubleshooting qualifications are described in historical material; issuer records and current status remain to be documented.

Record Notes

* Dates: the later electrical role is listed as 2022-2025 in supplied resumes; a 2025 employer wage-statement photo supports employment that year only. Earlier Alpha work is listed as 2010-2015 in the detailed historical account and 2010-2016 in later resumes. Exact dates remain unresolved.

Historical duties and training reflect supplied career accounts; public software and manuscript artifacts provide direct portfolio evidence. This master resume does not claim a PhD, Westminster enrollment, current credential validity, accepted journal publications or independently validated physical breakthroughs.

Education updated October 26, 2026; other historical record notes retained.